

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and maintained standardized, repeatable **Python** and **SQL** analysis pipelines for processing device characterization data, test & verification datasets, reliability and yield metrics, reducing manual data wrangling time by **40%**.
- Developed **Power BI** dashboards visualizing trends, pass/fail behavior, and performance metrics for engineering and manufacturing teams, enabling quicker identification of quality deviations.
- Applied **statistical methods** including trend analysis and variance analysis to support engineering decision-making on complex device performance data.
- Collaborated with engineering and manufacturing teams to define key metrics and KPIs for device quality and yield, ensuring data alignment across product development stages.
- Supported analysis of production and yield data, identifying trends and contributing to root cause analysis for improvement opportunities in manufacturing processes.
- Documented data models, assumptions, and analysis outputs in a structured format to support review, audit, and reuse by technical stakeholders.
- Containerized data analysis workflows using **Docker** and deployed them to **AWS EC2** instances, ensuring reproducible environments for test and validation data processing.

SKILLS

Data Analysis & Statistics – Python, SQL, R, Statistics, Trend Analysis, Variance Analysis, Anomaly Detection, Hypothesis Testing, Exploratory Data Analysis, A/B Testing, Time-Series Analysis

Business Intelligence & Visualization – Power BI, Tableau, DAX, Power Query, Data Modelling, Dashboards, Reporting

Programming & Scripting – Python, pandas, NumPy, scikit-learn, Java, Spring Boot, Flask, FastAPI

Databases & Data Warehousing – SQL, PostgreSQL, MySQL, Oracle, Data Warehousing, ETL

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, Git, Jenkins

Tools & Methodologies – Jira, Confluence, Agile, Scrum, Change Management, Data Traceability, Reproducibility

PROJECTS

Device Performance Analytics Platform

- Engineered **Python** data pipelines to ingest and process terabytes of device test data from multiple sources, standardizing formats for analysis.
- Developed interactive **Power BI** dashboards to visualize key performance indicators, failure modes, and reliability trends for engineering teams, enabling faster debugging cycles.
- Implemented **SQL** queries and data models to support ad-hoc analysis requests and validation of critical device performance metrics.

Manufacturing Yield Optimization Analysis

- Built automated **Python** scripts to extract and clean yield data from manufacturing execution systems, improving data accuracy by **15%**.
- Applied **statistical analysis** techniques to identify root causes of yield deviations, contributing to a targeted process improvement initiative that reduced defect rates by **8%**.
- Created a **SQL** data mart for efficient querying of historical yield performance, supporting trend analysis and predictive modelling efforts.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)